

TradeAlerts Simulation Engine Overview

Project Overview

TradeAlerts is a real-time simulation engine designed for evaluating technical stock alerts based on user-defined indicators and trading rules. It mimics real-world trading by enforcing market hours, fund settlement delays, and wash sale rules. The engine fetches live prices via the E*TRADE API and evaluates technical indicators on 1-minute historical bars. It is built to allow users to fine-tune entry/exit logic, simulate cash balances, and validate strategies with realistic constraints before going live with actual trades.

Supported Technical Indicators

Simple Moving Average (SMA)

The SMA is the average price over a given number of time periods. A bullish signal occurs when the current price is above the SMA, suggesting upward momentum.

Relative Strength Index (RSI)

RSI measures the magnitude of recent price changes to evaluate overbought or oversold conditions. A bullish signal is typically generated when RSI drops below a threshold (e.g., 30) and reverses upward.

RSI Slope

This measures the direction of the RSI line itself. A positive slope suggests building strength, reinforcing bullish sentiment.

MACD Histogram

The MACD Histogram is the difference between the MACD line and its signal line. A histogram crossing above zero indicates bullish momentum may be forming.

Bollinger Bands Breakout

This strategy checks whether the current price has broken above the upper Bollinger Band, which can signal the start of a strong trend or breakout.

Volume Spike

An optional filter to confirm moves with volume that exceeds a multiple of the average (e.g., 2x 5-day average). It strengthens signals by requiring heavier trading activity.

VWAP (Volume Weighted Average Price)

VWAP represents the average price weighted by volume throughout the trading day. If the current price is above VWAP by a threshold, it can suggest institutional buying pressure.

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Risk Management and Simulation Controls

The simulation includes multiple risk management options:

- Stop-Loss: Exit trade if price falls a fixed % below entry.
- Take-Profit: Exit trade if price exceeds a fixed % gain.
- Trailing Stop: Dynamically trails peak price and exits on pullback.
- Timed Exit: Optionally close position after N days.
- Single Entry Guard: Prevents re-entering a stock while already held.
- Wash Sale Filter: Prevents repurchase if a recent losing trade was made.
- Fund Settlement Delay: Prevents reuse of proceeds from sales until settlement completes (e.g., T+1).

Market Awareness

The engine respects real market open hours (9:30 AM to 4:00 PM ET) and uses NYSE calendar data to detect trading holidays and weekends. When markets are closed, it sleeps and resumes scanning when appropriate.